

2023 AMC 10B Problem 1 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10B Problem 1. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10B Problem 1

Problem:

Mrs. Jones is pouring orange juice into four identical glasses for her four sons. She fills the first three glasses completely but runs out of juice when the fourth glass is only $\frac{1}{3}$ full. What fraction of a glass must Mrs. Jones pour from each of the first three glasses into the fourth glass so that all four glasses will have the same amount of juice?

Answer Choices:

A. $\frac{1}{12}$

B. $\frac{1}{4}$

C. $\frac{1}{6}$

D. $\frac{1}{8}$

E. $\frac{2}{9}$

Solution:

Mrs. Jones has a total of $3\frac{1}{3} = \frac{10}{3}$ glasses of orange juice. Distributing the orange juice equally among the four glasses means that each glass receives $\frac{1}{4} \cdot \frac{10}{3} = \frac{5}{6}$ of a glass of orange juice. It follows that (C) $\frac{1}{6}$ of a glass of juice must be removed from each of the first three glasses.

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